

Designing VR training systems for children with attention deficit hyperactivity disorder (ADHD)

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ABSTRACT

Attention-deficit hyperactivity disorder (ADHD) is a common mental disorder in childhood, with a reported 5% global prevalence rate. The project uses Virtual Reality (VR) technology to help children improve their concentration in order to mitigate some of the various deficiencies in existing rehabilitation methods. The research aims to apply the interactive features of VR technologies and to combine them with psychological rehabilitation training technology. The research also uses Electroencephalography (EEG) brain electricity image technology for real-time information feedback. The mobile application can receive the EEG data with visualization to assist medical staff and patients' families in evaluating the treatment. The research designs a therapy training system without physical space restriction. It is easy to deploy and can be a highly customizable rehabilitation process.

Keywords: Virtual Reality, ADHD, EEG, interactive.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality, Mixed/augmented reality

1 INTRODUCTION

Attention deficit hyperactivity disorder (ADHD) manifests as a short attention span and hyperactivity [1]. Patients may be more impulsive than their neuro-typical peers in age and developmental level, and this is often accompanied with learning difficulties, behavioral disorders and maladjustment [1]. ADHD begins in childhood, but it can still show symptoms in adolescence and adulthood [2]. If the child does not receive suitable treatment for ADHD, there is an increased probability that this leads to several negative consequences in early life, such as difficulties at school and in the family context [2].

In China, parents of patients often deny mental disorders and resist psychiatric treatment [3]. The parents believe that diagnosing children as having neurological disorders is unacceptable. This often results in children missing the best opportunity for effective treatment. Thus, it is necessary to

encourage the family to accept the treatment. However, using pharmacological options to treat hyperactivity disorder creates concern over possible side-effects [4]. Consequently, the pharmacological options' side-effects are perceived to be potentially damaging to the children's overall health and therefore not the best primary treatment option.

Existing medical institutions usually run the supporting psychological rehabilitation to train the patients [4]. This training includes semi-structured group conversations, group or pair work, theme-related homework, and self-reflective inquiries [5]. The training usually uses physical and chemical factors to guide children to ignore the interference of foreign objects to concentrate on completing a single task [5] [6]. The patient's concentration can gradually increase through effective arrangement and appropriate periodic training during the treatment [6]. This program can help patients gradually overcome this psychological barrier to achieve the purpose of treatment. Although the training has positive effects, it also has some drawbacks. For instance, the pair work (patients) has time and space restrictions. Furthermore, the doctors cannot track the theme-related homework in real-time. The patients may ask parents for assistance to complete the task. Various restrictions may bring negative effects in the treatment. Therefore, it is necessary to discuss how to use information technologies to solve the restrictions.

VR technologies provide innovative Information and Communication Technologies (ICTs) to improve user-experience and environment interaction [7]. The technology opens a new horizon in the game industry. It also shows enormous potential in the educational, commercial, and medical fields without additional physical limitation [8]. Hansey designed a 3D virtual classroom with optional distraction factors to validate their effectiveness and convenience in ADHD diagnosis [9]. Romero ran a 125 participants group test in analyzing the effectiveness of VR-based interventions on cognitive deficits in children with ADHD [10]. After conducting extensive research on the VR field, we realized that VR technology can play a very important role in ADHD medical care. We propose a set of real-time tracking solutions by VR technologies. We design a feedback training system to improve children's ADHD treatment.

2 FRAMEWORK OVERVIEW

The VR interactive psychological rehabilitation training system with entertainment elements can provide a gamified learning environment to provide treatment without the, sometimes negatively perceived, connotations of medical treatment. It can potentially reduce the skepticism of parents and children during the treatment process. The system can assist the medical staff and patient's families in monitoring the patient's circumstances in real-time. Therefore, medical rehabilitation institutions can customize training for better training. The system can meet the following requirements:

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1. Use VR technology to create a good rehabilitation space for immersive experience.
2. No restriction by physical space and time for the rehabilitation training.
3. Real-time human-computer interaction feedback.

The system should include a VR rehabilitation treatment system and a brain wave sensing headset for data measurement based on the above design criteria. The following content describes how the research develops the system from both aspects.

2.1 Design Concept

The research aims to build the exploration flow concept. Compared to other ADHD VR projects, we do not only focus on building a single interactive environment, but also design a smart measurement flow to assist doctors and patients' families to monitor the patients' circumstances in real-time. Figure 1 illustrates the exploration flow concept in the VR Training System. The children can explore the VR training system at home without time and venue restrictions. The system simulates the interactive environments, including indoor and outdoor. The children need to complete the concentration tasks under the distraction elements. The EEG sensors measure the participant's physical data. The mobile application can visualize various data to doctors and patients' families in real-time. The doctors can customize the rehabilitation process on time.

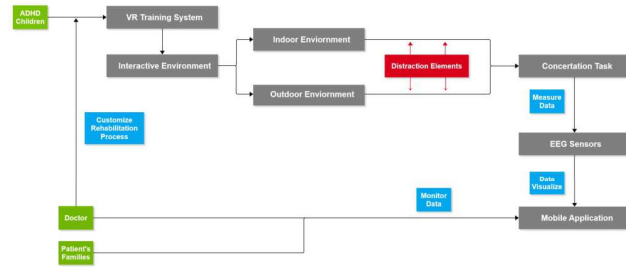


Figure 1: Exploration Flow Concept.

2.2 System Development

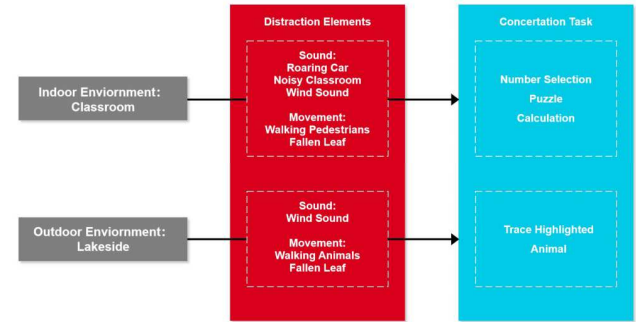
During the development of the system, the project uses modeling software such as 3dsMax and ZBrush to build models such as scenes, characters, and props. The model is topologically and textured by Substance Painter. After integrating various resources, the real-time rendering Unreal Engine 4 engine is applied to build the system interaction scene. According to the specific functions, the project uses C++ and blueprints to establish interactive functions.

2.3 Interactive Environment

In the existing literature, most existing ADHD VR projects use a single venue like the virtual classroom [9][11]. However, adapting to outdoor living is also an essential goal for ADHD children. Thus, this research simulates the outdoor and indoor environment.

Figure 2(a) illustrates the workflow in an interactive environment. The indoor environment is a learning environment: a typical classroom, with school-age children. In the classroom, the project simulates various factors that can cause distractions, such as a roaring car, walking pedestrians, a noisy classroom, the sound of wind and leaves outside the window (Figure 2(b)). The system guides children to complete some operations in the environment. The children first complete the task to select the correct number under a single distraction element (Figure 2(c)). With the training program moving on, there are more distraction

elements. The children complete more complicated tasks like puzzles and calculations. The gradual training process aims to reduce the influence of external interference factors on a child's concentration. The interaction mechanism can assist children to stay focused in their daily learning and life.



(a)



(b)



(c)

Figure 2: Interactive Environment: (a) Working Flow; (b) Distraction Elements (Indoor); (c) Concentration Task (Number Selection).

The outdoor environment is the "Lakeside." The system allows children to explore the environment and trace with the highlighted animal. Compared to the indoor environment, the "Lakeside" provides a beautiful scene, which can potentially decrease children anxiety. The tracing task is simple and encourages children to participate in and concentrate on it.

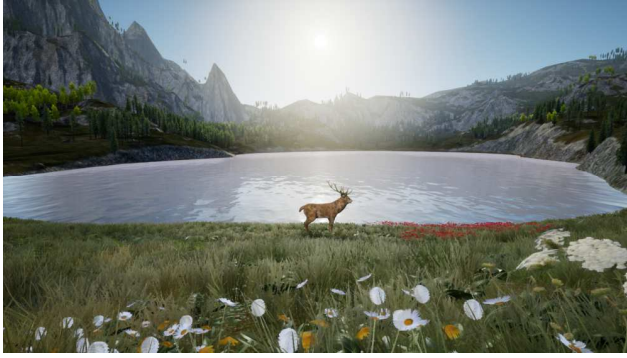


Figure 3: Outdoor Environment.

2.4 Data Measurement

When the children explore the environment, data measurement is essential to capture the physiological measures. The research uses EEG sensors to measure the data. The EEG brain wave sensing headset is an EEG device containing dry electrodes, TGAM core PCB modules, and Bluetooth sensors [12]. The device collects the voltage difference on the scalp through the forehead electrode to collect brain wave data [12]. The TGAM core can collect the original EEG signal and then send the data signal to the computer or mobile phone through the Bluetooth sensor [12].

Three-electrode pervasive EEG collector has high accuracy from 78.24% to 91.25% [13][14]. Jian analyzed that the average accuracy reaches 83.07% [15]. Besides, the collector price is affordable. Thus, our research uses the three-electrode pervasive EEG collector as shown in Figure 4(a).

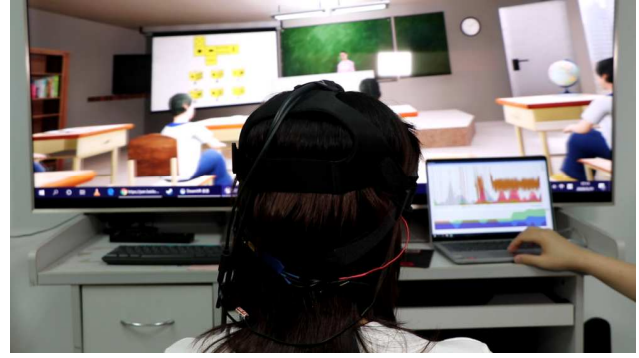
We group the VR device and the EEG headset together. We allow the user to wear the VR device (HTC Vive) to explore the environment (Figure 4(b)). While the user starts to play, the computer can receive the related data (Figure 4(c)).

The team can directly see the data feedback from the software (Mind Viewer) and know the state of the device user in real-time (Figure 4(b)). Mind Viewer provides the essential parameters: Raw Wave, Delta, Theta, Alpha, Beta, and Gamma, as shown in the right orange wireframe in Figure 4(c). Numerous studies have examined that the ADHD patients show an increased Delta wave and Theta wave activities, decreased Alpha wave and Beta wave activities [9][16][17]. We can monitor the wave activities to measure the patient's status.

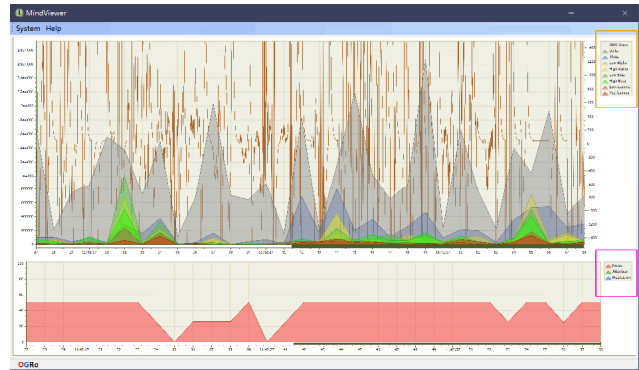
Mind Viewer has concluded Noise, Attention, and Meditation value by analyzing essential parameters. The right purple framework in Figure 4(c) shows these parameters in real-time. When Noise is the highest and other parameters have no value as Figure 4(c), wear the sensor correctly is necessary. The Noise data visualization guide us to filter noise in time.



(a)



(b)



(c)

Figure 4: Data Measurement by EEG: (a) EEG Collector; (b) User Test Session; (c) Data collection.

2.5 Mobile Application

Although in the described VR system, data is received through an EEG sensor, there is no means to provide patients and the medical staff with real time data visualization. Therefore, the project is developing a mobile application that allows patients and medical staff to receive the data and adjust the treatment on time.

The research has built the application's concept structure and designed the initial user interface prototype. Users can receive EEG information more quickly and conveniently through the application and make corresponding adjustments in real-time according to their feedback. The main modules include Status, My Device, Real-time Data, Now Playing, and Add Device (Figure 5). The Real-time Data module displays the data of Attention and Meditation from Mind Viewer.

The device connection method can be provided immediately when the user opens the APP. After the device is connected, the main content can be briefly presented. Combined with the above content, the state changes of users can be observed in real-time, which is convenient for finding suitable treatment plans.

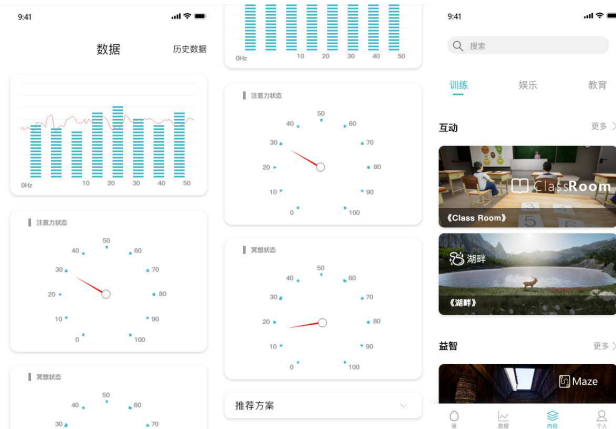


Figure 5: Mobile Application Prototype.

3 CONCLUSION

Existing ADHD rehabilitation training mainly relies on physical scenes, with disadvantages such as high communication cost, low treatment efficiency, and difficulty in replicating physical scenes. VR technology can solve the existing psychological rehabilitation therapy limitations due to space and physical entity factors. It provides a new direction for various psychological rehabilitation training programs. The EEG headset can amplify and record the spontaneous biopotentials of the brain from the scalp to obtain graphical data, which can quantify feedback on the wearer's concentration. By performing computer simulation and receiving sensor feedback, it is possible to precisely adjust and control parameters such as materials and task complexity presented in training in a virtual 3D environment. According to the real-time data feedback and the recorded training process, rehabilitation strategies and personal adaptation can be customized. Consequently, the entire training can be scientifically and accurately quantified.

4 FUTURE WORK

Although we have built the VR system and discussed how to measure the data, we have not recruited a large group of users to start the formal experiment. Furthermore, the mobile application is still in the initial developing stage. In the future, we will run a large group of user tests to receive the feedback to optimize the interactive functions through user feedback. We will continue developing the mobile data feedback analysis in the mobile application. We hope that the research can contribute to the VR training field.

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